

SAMUEL H. RUSSELL

Principal Data Architect

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PROFESSIONAL SUMMARY

Data architect with 15+ years on both ends of enterprise data: the Oracle systems that run the business and the Databricks lakehouses that analytics and AI teams build on. At Fiserv, built GoldenGate CDC feeds into Delta Lake while live financial transactions kept flowing, and cut platform infrastructure costs 60% by moving multi-terabyte workloads from Oracle to Databricks and Snowflake. Founded Sightline Signals and, separately, is the sole inventor on 6 U.S. provisional patent applications in AI agent infrastructure and data center energy optimization. Hands-on in Python, SQL, PL/SQL, and PySpark, mostly in regulated financial, federal, and defense environments on AWS, AWS GovCloud, Azure, and OCI, and holds an active Secret clearance.

CORE EXPERTISE

Databricks, Lakehouse, and Streaming: Delta Lake (Liquid Clustering, Z-Ordering, Time Travel, Deletion Vectors), Unity Catalog, Delta Live Tables, DLT Expectations, Auto Loader, Databricks Workflows, Asset Bundles, Databricks CLI, Databricks SQL Serverless, Lakehouse Monitoring, System Tables, medallion architecture, dimensional and data vault modeling, Great Expectations, dbt, Apache Airflow; PySpark, Spark SQL, Structured Streaming with watermarking and exactly-once processing, AQE, Catalyst Optimizer, broadcast and shuffle join tuning, data skew remediation, memory and spill profiling; Kafka, Azure Event Hubs, MQTT and LoRaWAN ingestion; time-series architecture, device and asset identity modeling, high-cardinality retention, building management and facility telemetry, data-quality classification for stale, duplicate, and out-of-order data; Power BI, Tableau

Oracle and Database Platforms: Oracle Database 10g, 11g, 12c, 19c, and 21c; RAC; Grid Infrastructure; Clusterware; ASM; CDB and PDB multitenant; E-Business Suite 12.1 to 12.2; Oracle Supercluster; OPatch and rolling RAC patching; Oracle Enterprise Manager; AWR, ADDM, ASH, SQL Plan Management and SQL Plan Baselines, SQL Profiles, SQL Tuning Advisor, execution plan diagnostics with DBMS_XPLAN, wait event and DB Time analysis, CBO and SQL trace analysis with TKPROF, Linux kernel tuning (HugePages, RAC private interconnect); PostgreSQL, Snowflake, MongoDB, DB2, TimescaleDB, SQLite

Availability and Recovery: Data Guard, Data Guard Broker, Active Data Guard, Fast-Start Failover, Far Sync, RMAN active duplication and block media recovery, Flashback, point-in-time recovery, high availability and disaster recovery design, live switchover drills

Migration, Replication, and Integration: near-zero downtime cutovers, rollback plans, migration run books, database consolidation; Oracle GoldenGate Microservices CDC (Extract, Replicat, trail encryption) for live cutovers; Change Data Feed, Delta MERGE, SCD Type 1 and Type 2; Data Pump, Transportable Tablespaces; Oracle to Databricks, Snowflake, and PostgreSQL; DB2 on Windows to Oracle RAC on Red Hat Linux; legacy Unix to Oracle Supercluster; AWS DMS, AWS SCT, Azure Database Migration Service; row-count and checksum reconciliation; REST APIs

AI Data Platforms and Tools: Mosaic AI Vector Search (Delta Sync and hybrid ANN search), Model Serving (Foundation Model APIs, provisioned throughput), Mosaic AI Agent Framework, MLflow 3.x tracing, evaluation, and judge models, Model Registry, Feature Store, compound AI systems, RAG data pipelines, batch and streaming embedding generation, chunking strategy, vector index governance in Unity Catalog, groundedness and hallucination evaluation, A/B testing; Anthropic Claude and Claude API, OpenAI API and Codex, Google Gemini and Google Antigravity; tiered LLM routing with fallback, prompt caching, agent memory

Governance and Security: Unity Catalog fine-grained permissions, row and column security, dynamic masking, data lineage, TDE with keystore and wallet management, Database Vault, Unified Auditing, TLS enforcement, default-deny egress, hash-chained signed audit trails, AWS KMS, Azure Key Vault, FISMA and ATO, FedRAMP, NIST 800-53, NERC CIP, CIS Benchmark Level 1, DoD cybersecurity directives, DISA STIGs

Cloud, DevOps, and Programming: AWS (S3, Glue, EMR, Redshift, Lambda, IAM, EC2), AWS GovCloud, Azure, OCI, Terraform, CloudFormation, ARM and Bicep, Docker, Kubernetes, VPC and VNet peering, Direct Connect, ExpressRoute, VPN; GitHub Actions, Azure DevOps, Jenkins, Git, CI/CD release gates (testing, static analysis), Ansible, CloudWatch, Azure Monitor, Datadog, Splunk, ServiceNow, ITIL; Linux (RHEL 8 and 9, OEL), Unix, Windows Server; Python, SQL, PL/SQL, Bash, JSON Schema validation

Domains: 24/7 financial transaction processing, NASA Artemis on AWS GovCloud, defense logistics ERP (GCSS-MC, GCSS-Army PBUSE), AI data centers and multi-site facility portfolios

PROFESSIONAL EXPERIENCE

Founder and Principal Cloud Data AI Architect

Sightline Signals | November 2024 - Present

Sightline Signals adds a read-only analytics layer on top of the BMS already installed on-premise in AI data centers, warehouses, and multi-site portfolios. It uses the telemetry those systems already collect: HVAC, power, water, IT, access control, and fire and life safety.

- Founded the company and lead a 6-person team of co-founders and one other engineer, owning the platform's design and architecture.
- Built the production GenAI diagnostic platform, covering ingestion, vector pipelines, agent reasoning, security, CI/CD, APIs, and the delivered application.
- Built Delta Lake and Spark Structured Streaming pipelines (DLT Expectations, Auto Loader) for high-cardinality sensor telemetry, with automated schema evolution and a quarantine path for bad data.
- Implemented a RAG diagnostic engine on Mosaic AI Vector Search and Model Serving, using Unity Catalog Functions for governed tool calling against telemetry, equipment manuals, and incident history.

- Added MLflow tracing and evaluation to the RAG and agent workflows to measure retrieval precision, groundedness, and hallucination rate. That work helped cut inaccurate diagnostic output by 58%.
- Delivered close to 35 screens: ranked operating risk and remediation guidance for facility teams, and energy cost, savings, and ESG reporting for executives, each traceable to source data.

Oracle OCI RAC Senior DBA/Migration Expert

Fiserv | July 2022 - November 2024

- Moved multi-terabyte analytical workloads from Oracle to Databricks Delta Lake and Snowflake, cutting platform infrastructure costs by 60%.
- Built GoldenGate CDC pipelines into Databricks using Change Data Feed and Delta MERGE while financial systems kept processing live transactions.
- Applied Unity Catalog row- and column-level security, dynamic masking, and audit lineage to regulated financial datasets.
- One of three architects leading technical delivery for a 7-person data engineering, compliance, and platform team on a system that supports 24/7 financial processing.

Database Engineer

NASA Artemis Program | June 2021 - July 2022

- Ran 6 Oracle, PostgreSQL, Snowflake, and MongoDB databases supporting Artemis telemetry and mission planning in AWS GovCloud under FedRAMP and FISMA requirements.
- Advised on new software and major updates as part of the team that maintained the environment's Authority to Operate (ATO).
- Migrated 5 TB of program analytics from Oracle to Snowflake and kept the environment audit-ready the whole way.
- Automated vulnerability scanning, configuration management, and patching with Python, which cut manual effort by around 40%.

Senior Cloud DBA Applications Technical

GM Financial | May 2019 - June 2021

- Led a 3-person team upgrading 70+ Oracle RAC databases (about 18 TB) from 12c to 19c across AWS and Azure, inside 5-hour Saturday maintenance windows.
- Handled the technical side of the Oracle E-Business Suite 12.1 to 12.2 upgrade, working alongside the functional lead.
- Wrote Python, PL/SQL, and Bash automation that cut manual database administration work by roughly 25%.

Senior Oracle Application SME

NAVWAR, United States Navy | February 2015 - May 2019

- Oracle subject matter expert for the Marine Corps GCSS-MC ERP upgrade from E-Business Suite 12.1 to 12.2, a multi-year program serving 16,000 users across 62 bases worldwide.
- Set technical direction for both government staff and contractor teams, and answered for the database layer during compliance reviews.
- Led a database consolidation that reduced the server footprint and tightened security.
- Kept the ERP databases behind Marine Corps logistics reliable and in line with federal regulations and DoD cybersecurity standards for the length of the upgrade.

Senior Oracle DBA

McLane Advanced Technologies | August 2013 - February 2015

- Supported Oracle database implementation projects for several federal and state government clients at once.
- One of 2 Oracle DBAs on a 6-person team that upgraded GCSS-Army PBUSE from a legacy Unix platform to Oracle Supercluster, cutting inventory reporting response time 60% at the local Army base and deployed depots.
- Served as lead architect representative in the architecture review for the Texas Railroad Commission's legacy modernization and helped shape the plan.

Senior RAC Oracle DBA

Oracle Corporation | August 2010 - August 2013

- Hired as the production RAC conversion expert, the first hire from outside Siebel, to move Siebel from Windows and DB2 onto Red Hat Linux and Oracle.
- Led the migration from DB2 to Oracle 10g RAC, using GoldenGate to replicate data in real time.
- One of 4 engineers sent to evaluate GoldenGate as the replication solution for the conversion.

CERTIFICATIONS AND EDUCATION

Certifications: Databricks Certified Generative AI Engineer Associate; Oracle Database 12c Certified Professional; CompTIA Security+ (DoD 8570 and 8140 compliant); Microsoft Certified Technology Specialist

Education: Bachelor of Science in Management Information Systems, University of Oklahoma

Patents: Sole inventor on 6 U.S. provisional patent applications covering AI agent infrastructure (tiered LLM routing with fallback synthesis, prompt-cache partitioning, query-scored capability injection, agent memory) and closed-loop data center energy optimization

Security Clearance: Active Secret

Military Service: United States Air Force, Air Surveillance Technician and Instructor